Cardiovascular System (CARD) · NPLEX-P1-CARD-15

CV-15 Heart and Lung Interactions

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-15-01-C01-BD1

Brain dump 1 · foundational retrieval: Build the two circuits side by side

A competent response contains

· The two circuits are in series, so each ventricle ejects about 5 L/min at rest and any pressure difference between them reflects a difference in resistance.

· Pulmonary artery pressure is about 25/8 mmHg with a mean near 15 mmHg; systemic arterial pressure is about 120/80 mmHg with a mean near 93 mmHg.

· Left atrial pressure is about 5 mmHg and right atrial pressure about 2 mmHg.

· Pulmonary capillary hydrostatic pressure is about 7 mmHg, with some sources giving about 10 mmHg; systemic capillary pressure is about 17 mmHg.

· Resistance at 5 L/min is about 2 Wood units in the pulmonary circuit and about 18 Wood units in the systemic circuit, a Wood unit being 1 mmHg per L/min.

· The pulmonary pressure drop is about 10 mmHg against about 90 mmHg systemically, so with equal flow the resistance is about one ninth.

· Pulmonary resistance is spread across arteries, capillaries, and veins; systemic resistance sits mainly in the arterioles.

· Pulmonary artery walls are about one third as thick as the aorta's and pulmonary arterioles have little smooth muscle, so caliber is set mainly by the pressures inside and around the vessels.

· The pulmonary circuit holds about 450 mL of blood and is highly compliant.

· Bronchial arteries from the aorta supply the airways at systemic pressure and carry about 1 to 2% of cardiac output.

· Low pressure has two consequences: the thin right ventricle tolerates a sudden rise in afterload poorly, and a capillary pressure far below the plasma oncotic pressure of about 28 mmHg keeps the alveoli dry.

· Pulmonary hypertension is a sustained resting mean pulmonary artery pressure above about 20 mmHg.

Distinctions that matter

· Flow is the same in both circuits; only the pressure drop and therefore the resistance differ.

· Bronchial arteries supply lung tissue at systemic pressure, while pulmonary arteries carry the whole cardiac output at low pressure for gas exchange.

Common misconceptions

· Writing that the pulmonary circuit carries less blood flow than the systemic circuit because its pressures are lower. The circuits are in series and carry the same output.

· Placing pulmonary resistance in the arterioles by analogy with the systemic bed. Pulmonary arterioles have little muscle and resistance is distributed across the whole bed.

Where it is taught: Section C01, first paragraph, the pressure table, and the paragraph on consequences of low pressure of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C01-BD2

Brain dump 2 · integration and application: Work the pulmonary resistance numbers through exercise and disease

A competent response contains

· At rest, pulmonary vascular resistance is (15 minus 5) divided by 5 L/min, which is 2 Wood units.

· In exercise it is (22 minus 10) divided by 15 L/min, which is 0.8 Wood units, so flow tripled while resistance fell by more than half.

· Had resistance stayed at 2 Wood units, mean pulmonary artery pressure would have approached 40 mmHg.

· The two passive mechanisms are recruitment, the opening of capillaries that were closed or unperfused at rest and mostly apical, and distension, the widening of capillaries already open.

· Recruitment dominates at low pressure and distension at higher pressure.

· Resistance varies inversely with the fourth power of the radius, so modest widening lowers resistance greatly.

· Capillary transit time falls from about 0.8 s to about 0.3 s, which is still enough for oxygen to equilibrate.

· Together these mechanisms spare the right ventricle, limit filtration, and enlarge the gas exchange surface.

· At high lung volume the alveolar vessels, the capillaries in the alveolar walls, are stretched and compressed, so resistance rises, as in hyperinflation or high positive end-expiratory pressure.

· At low lung volume the extra-alveolar vessels lose radial traction and narrow, as in atelectasis; they are pulled open as the lung expands.

· Total pulmonary vascular resistance is therefore lowest near functional residual capacity and rises at both extremes.

· Recruitment and distension are a reserve. After removal of one healthy lung the remaining bed recruits and distends, so resting pressure rises little.

· When bed is lost to emphysema, emboli, or arterial remodeling, the remaining vessels are already recruited at rest, so further flow raises pressure steeply and the first signs are pulmonary hypertension and dyspnea on exercise.

Distinctions that matter

· Recruitment adds parallel channels; distension widens existing ones. Both are passive responses to pressure, not active dilation.

· Alveolar and extra-alveolar vessels respond to lung volume in opposite directions, which is why the resistance curve against lung volume is U shaped.

Common misconceptions

· Attributing the fall in resistance during exercise to vasodilator release. The pulmonary bed has little smooth muscle and the change is passive.

· Expecting resting pulmonary artery pressure to reveal early pulmonary vascular disease. The reserve hides it until exercise uses the reserve up.

Where it is taught: Section C01, paragraph on why resistance falls as flow rises, the worked example, and the paragraphs on lung volume and lost reserve of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C02-BD1

Brain dump 1 · foundational retrieval: Draw hypoxic pulmonary vasoconstriction from sensor to constriction

A competent response contains

· Hypoxia inhibits the voltage-gated K+ channels Kv1.5 and Kv2.1 that hold the resting potential of pulmonary artery smooth muscle negative.

· The membrane depolarizes and L-type Ca2+ channels open.

· Calcium enters through those channels and is also released from the sarcoplasmic reticulum, activating calmodulin and myosin light chain kinase.

· Rho-kinase inhibits myosin light chain phosphatase and so raises calcium sensitivity.

· Endothelial nitric oxide falls and endothelin-1 rises, adding to the constriction.

· The stimulus is mainly alveolar PO2, which diffuses directly to the small muscular arteries, and the response becomes strong below about 70 mmHg.

· The response persists in an isolated denervated lung, which shows it is local rather than neural.

· Its purpose is to divert blood away from poorly ventilated alveoli toward better ventilated lung, preserving ventilation-perfusion matching, since blood sent to an unventilated alveolus is wasted.

· The upstream oxygen sensor is debated; the leading proposal is a change in mitochondrial reactive oxygen species and redox state.

Distinctions that matter

· Alveolar PO2 rather than arterial or mixed venous PO2 is the main stimulus, because it reaches the small arteries by direct diffusion.

· Calcium entry raises calcium concentration while Rho-kinase raises sensitivity to the calcium already present; both contribute to the same contraction.

Common misconceptions

· Saying hypoxia opens K+ channels in these cells. It inhibits them, and inhibition is what depolarizes the membrane.

· Treating the response as reflex or sympathetic. An isolated denervated lung still constricts.

Where it is taught: Section C02, opening paragraph, the paragraph on ion channel events, and the hypoxic pulmonary vasoconstriction flow diagram of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C02-BD2

Brain dump 2 · integration and application: Regional against global hypoxia, and what chronic hypoxia leaves behind

A competent response contains

· Pneumonia is regional hypoxia: constriction diverts blood to healthy lung, limits the shunt, and protects arterial oxygen, while mean pulmonary artery pressure barely changes because the rest of the bed recruits.

· Rapid ascent to 4,000 m is global hypoxia: every region constricts at once, pulmonary vascular resistance rises, and nothing is gained in matching.

· Long-standing COPD with chronic hypoventilation is also global, with hypoxia plus loss of capillary bed from emphysema, giving mild to moderate pulmonary hypertension and, over years, chronic cor pulmonale.

· Hypercapnia and acidosis strengthen the constriction.

· On rapid ascent, constriction is uneven, so less constricted regions are overperfused at high pressure, which is the accepted explanation for high-altitude pulmonary edema.

· Sustained hypoxia stabilizes hypoxia-inducible factors HIF-1 alpha and HIF-2 alpha.

· These drive medial hypertrophy, muscularization of arterioles that normally lack muscle, and intimal and adventitial fibrosis, which add a fixed component to the reversible constriction.

· HIF also raises erythropoietin, and the resulting polycythemia raises viscosity and therefore resistance.

· In the fetus, constriction in the fluid-filled lung keeps pulmonary vascular resistance high and sends right ventricular output through the ductus arteriosus.

· At birth, lung expansion, rising alveolar PO2, and nitric oxide lower resistance sharply; failure of that fall is persistent pulmonary hypertension of the newborn.

Distinctions that matter

· Regional hypoxia uses the response as intended and barely raises pressure; global hypoxia raises resistance with no matching benefit.

· Constriction is reversible with oxygen, while HIF-driven remodeling and polycythemia are not, which is why chronic pulmonary hypertension becomes fixed.

Common misconceptions

· Expecting mean pulmonary artery pressure to rise substantially in lobar pneumonia. The unaffected bed recruits and absorbs the diverted flow.

· Treating high-altitude pulmonary edema as a left heart problem. It follows uneven constriction and overperfusion of the remaining open regions.

Where it is taught: Section C02, paragraphs on regional and global hypoxia and on remodeling, and the stop and predict box of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C02-RT01

Retrieval exercise: Rebuild the hypoxic pulmonary vasoconstriction sequence

A competent response contains

· Step 1: regional alveolar PO2 falls below about 70 mmHg.

· Step 2: Kv channels in pulmonary artery smooth muscle are inhibited; while open they carry K+ outward and hold the resting potential negative.

· Step 3: the membrane depolarizes and L-type Ca2+ channels open.

· Step 4: calcium from those channels and from sarcoplasmic reticulum release drives contraction, with Rho-kinase raising calcium sensitivity by inhibiting myosin light chain phosphatase.

· Step 5: the small arteries and arterioles of that region constrict.

· Step 6: blood is diverted to ventilated lung, preserving ventilation-perfusion matching.

· In a hypoxic systemic bed the opposite happens: adenosine acts on A2 receptors, falling ATP opens ATP-sensitive K+ channels, the membrane hyperpolarizes, and the vessel dilates, with lactate, H+, CO2, nitric oxide, and prostacyclin adding to the dilation.

Distinctions that matter

· Both beds use K+ channels, but closing Kv channels depolarizes pulmonary muscle while opening KATP channels hyperpolarizes systemic muscle.

· The pulmonary response matches blood to ventilated alveoli; the systemic response matches blood to tissue oxygen demand.

Common misconceptions

· Drawing the systemic vessel as constricting as well. Systemic arterioles dilate in hypoxia, which is the whole point of the contrast.

Where it is taught: Section C02, flow diagram on hypoxic pulmonary vasoconstriction and the comparison table of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C02-CP01

Comparison table: Pulmonary against systemic arterioles

A competent response contains

· Local hypoxia: pulmonary arterioles constrict, systemic arterioles dilate.

· Ion channel event: pulmonary Kv channels close and the membrane depolarizes; systemic KATP channels open and the membrane hyperpolarizes.

· Other mediators: pulmonary constriction gains from less endothelial nitric oxide, more endothelin-1, and Rho-kinase; systemic dilation gains from adenosine at A2 receptors, lactate, H+, CO2, nitric oxide, and prostacyclin.

· Purpose: the pulmonary response sends blood to ventilated alveoli, the systemic response sends blood to tissue that needs oxygen.

· Acidosis and a high CO2 tension constrict pulmonary arterioles, strengthening the hypoxic response, and dilate systemic arterioles.

· Global hypoxia gives pulmonary hypertension in the lung, and systemic vasodilation offset by reflex sympathetic tone in the body.

· Pulmonary arterioles have little smooth muscle and share resistance with arteries, capillaries, and veins; systemic arterioles are heavily muscled and carry most of the systemic resistance.

Distinctions that matter

· The direction of the membrane potential change is the quickest way to keep the two beds apart: depolarization in the lung, hyperpolarization in the tissues.

· Carbon dioxide and acidosis move the two beds in opposite directions as well, following the same logic as oxygen.

Common misconceptions

· Assuming any vessel dilates when its environment is hypoxic. In the lung, low alveolar oxygen is a signal that ventilation has failed, not that the tissue needs more flow.

· Treating the pulmonary bed as a resistance bed governed by arteriolar tone like the systemic bed.

Where it is taught: Section C02, table on pulmonary versus systemic arteriolar responses, with wall structure in section C01 of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C03-BD1

Brain dump 1 · foundational retrieval: One breath, four chambers, and the second heart sound

A competent response contains

· Transmural pressure is the pressure inside a chamber minus the pressure outside it, and the heart and great veins sit inside the pleural pressure, so one change in pleural pressure changes the filling pressure of every chamber at once.

· Pleural pressure is about -5 cmH2O at functional residual capacity and about -7.5 cmH2O at end inspiration.

· Venous return is driven by mean systemic filling pressure, about 7 mmHg, minus right atrial pressure.

· In spontaneous inspiration, right atrial pressure falls relative to the extrathoracic veins and the descending diaphragm raises abdominal pressure and pushes blood toward the thorax, so venous return and right ventricular stroke volume rise in the same breath.

· Left ventricular stroke volume falls slightly for three reasons: the expanded pulmonary vessels hold more blood, the fuller right ventricle bows the septum to the left, and the more negative pleural pressure slightly raises left ventricular afterload.

· The extra right ventricular output reaches the left heart a few beats later, in early expiration.

· P2 is delayed by the longer right ventricular ejection and by the low impedance of the expanded pulmonary vessels, which lets blood keep moving forward after right ventricular pressure starts to fall.

· A2 comes slightly earlier because left ventricular stroke volume is slightly smaller.

· The result is physiologic splitting of S2 in inspiration only, best heard at the left upper sternal border because P2 is soft.

· The normal inspiratory fall in systolic pressure is less than 10 mmHg.

Distinctions that matter

· The two ventricles respond to the same breath in opposite directions, and the lag between them is what the pulse and the heart sounds record.

· Ventricular interdependence is a mechanical effect through the septum and pericardium, separate from the volume the pulmonary vessels hold.

Common misconceptions

· Saying inspiration raises the output of both ventricles. Right output rises within the breath and left output falls slightly until the extra volume arrives.

· Explaining the split by saying A2 is delayed. A2 is slightly early and P2 is delayed.

Where it is taught: Section C03, first two paragraphs, the inspiration flow diagram, and the paragraph on P2 of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C03-BD2

Brain dump 2 · integration and application: Exaggerated and reversed breathing effects

A competent response contains

· Pulsus paradoxus is an inspiratory fall in systolic pressure greater than 10 mmHg, an exaggeration of the normal dip rather than a reversal of it.

· In tamponade, pericardial fluid fixes the total volume of the two ventricles, so inspiratory right ventricular filling pushes the septum into the left ventricle and left stroke volume falls far more than normal; the same compression equalizes diastolic pressures and explains hypotension and distended neck veins.

· In severe asthma or COPD the pericardium is normal, and very large pleural pressure swings raise right heart filling and left ventricular afterload instead. Massive pulmonary embolism and severe hypovolemia can also produce it.

· It is less common in constrictive pericarditis, where the rigid pericardium shields the heart from the pleural pressure change, and jugular venous pressure rises on inspiration rather than falling.

· A ventilator raises pleural pressure during inspiration and positive end-expiratory pressure keeps it raised, so right atrial pressure rises, the gradient for venous return narrows, and cardiac output falls.

· The fall is largest in hypovolemia, because mean systemic filling pressure is already low.

· High lung volumes also compress alveolar vessels and can create zone 1 lung.

· The raised pressure around the left ventricle lowers its afterload, so in a failing left ventricle with high filling pressures stroke volume may be maintained, while with low filling pressures the loss of preload dominates.

· The strain phase of a Valsalva maneuver acts the same way, lowering venous return and pulse pressure and triggering a baroreflex rise in heart rate.

· Physiologic splitting: split in inspiration only, because inspiration delays P2 and advances A2.

· Wide splitting: split in both phases and wider in inspiration, from delayed right ventricular emptying, as in right bundle branch block or pulmonic stenosis.

· Fixed splitting: a wide split unchanged by breathing, as in atrial septal defect, because shunt flow falls as venous return rises and right ventricular volume stays constant.

· Paradoxical splitting: split in expiration and single in inspiration, because A2 is delayed until after P2, as in left bundle branch block or severe aortic stenosis.

Distinctions that matter

· Tamponade works through a fixed pericardial volume; severe airflow obstruction works through the size of the pleural pressure swing.

· Spontaneous and positive pressure breathing move pleural pressure in opposite directions, so every downstream effect reverses.

Common misconceptions

· Reading the name pulsus paradoxus as a reversal of the normal pattern. It is the normal direction, exaggerated beyond 10 mmHg.

· Expecting positive pressure ventilation to lower cardiac output in every patient. With a failing left ventricle and high filling pressures the fall in afterload can hold stroke volume up.

Where it is taught: Section C03, pulsus paradoxus paragraph, positive pressure ventilation paragraph, and the S2 splitting table of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C03-RT01

Retrieval exercise: Rebuild the inspiration sequence through to S2

A competent response contains

· Step 1: pleural pressure falls.

· Step 2: venous return and right ventricular stroke volume rise, helped by the fall in right atrial pressure relative to the extrathoracic veins and by the descending diaphragm raising abdominal pressure.

· Step 3: right ventricular ejection lasts longer, delaying P2, with the low impedance of the expanded pulmonary vessels adding to the delay.

· Step 4: the pulmonary vessels hold more blood and the septum bows left.

· Step 5: left stroke volume falls slightly, so A2 comes slightly earlier; the three reasons are the pooled pulmonary volume, the septal shift, and the small rise in left ventricular afterload.

· Step 6: S2 splits and systolic pressure dips by less than 10 mmHg.

· In expiration each step reverses: pleural pressure rises, venous return and right stroke volume fall, P2 moves earlier, the pulmonary vessels hold less blood, left stroke volume rises, and S2 becomes single.

· The extra right ventricular output reaches the aorta a few beats later, in early expiration.

Distinctions that matter

· The split closes in expiration because P2 moves earlier, not because A2 moves later.

· Right and left stroke volume change within the same breath but not at the same time, and the lag is what produces the pressure dip.

Common misconceptions

· Putting the rise in left ventricular output in the same breath as the rise in right ventricular output. It arrives a few beats later.

Where it is taught: Section C03, flow diagram on inspiration, ventricular output, and S2 of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C04-BD1

Brain dump 1 · foundational retrieval: Starling forces at the pulmonary capillary and the safety margin

A competent response contains

· Outward forces: capillary hydrostatic pressure about 7 mmHg, interstitial hydrostatic pressure about -8 mmHg acting as suction, and interstitial oncotic pressure about 14 mmHg, totaling about 29 mmHg.

· Inward force: plasma oncotic pressure about 28 mmHg.

· Net outward filtration is about 1 mmHg, and the lymphatics carry that filtrate away.

· Three defences hold off edema as capillary pressure rises: increased lymph flow, loss of interstitial suction, and dilution of interstitial protein.

· Alveolar flooding begins once capillary pressure approaches plasma oncotic pressure of about 28 mmHg, so the safety margin is about 21 mmHg.

· Interstitial edema forms first and the alveoli flood second.

· A low plasma albumin lowers plasma oncotic pressure and narrows the margin.

· Chronic elevation widens the margin, with lymphatics able to enlarge about tenfold, so some patients with mitral stenosis tolerate a capillary pressure of 40 mmHg.

· Permeability injury, as in ARDS, lets protein cross the barrier, so protein-rich edema forms at a normal capillary pressure.

Distinctions that matter

· Hydrostatic edema comes from a raised capillary pressure with an intact barrier; permeability edema comes from an injured barrier at a normal pressure, and the protein content of the fluid separates them.

· Interstitial oncotic pressure in the lung is high enough to matter, which is why the pulmonary Starling balance is not simply capillary pressure against plasma oncotic pressure.

Common misconceptions

· Treating interstitial hydrostatic pressure as positive. In the lung it is negative and pulls fluid out of the capillary.

· Expecting the same capillary pressure to flood every patient's alveoli. The margin is narrowed by low albumin and widened by years of raised pressure.

Where it is taught: Section C04, paragraph on Starling forces and the safety margin of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C04-BD2

Brain dump 2 · integration and application: Which failing chamber, and which side of the circuit caused it

A competent response contains

· COPD is group 3 pulmonary hypertension from lung disease and hypoxia, through hypoxic constriction, remodeling, and loss of capillary bed, and the term cor pulmonale applies.

· Mitral stenosis is group 2, from backward transmission of left atrial pressure, and cor pulmonale does not apply because the cause is left heart disease.

· The five groups: 1 pulmonary arterial hypertension, including idiopathic, BMPR2, and connective tissue disease, with endothelial dysfunction and plexiform lesions; 2 left heart disease; 3 lung disease or hypoxia, including COPD, interstitial disease, and sleep apnea; 4 chronic thromboembolism with organized clots obstructing arteries; 5 multifactorial, for example sarcoidosis.

· Groups 1, 3, and 4 load the right ventricle from the lung side, so cor pulmonale applies to them.

· Cor pulmonale is right ventricular hypertrophy, dilation, or failure from pulmonary hypertension of lung, pulmonary vascular, or ventilatory origin, and right heart failure from left heart disease is excluded even though it is the most common cause of right heart failure overall.

· The right ventricle meets slowly rising afterload with concentric hypertrophy, since by the law of Laplace a thicker wall lowers wall stress.

· It later dilates, the tricuspid valve leaks, and systemic venous pressure rises, giving distended jugular veins, a nutmeg liver, and dependent edema.

· Pulmonary hypertension gives a loud P2 and a parasternal heave, and the ECG shows right axis deviation, a tall R wave in V1, and P pulmonale in lead II.

· In mitral stenosis, pulmonary capillary wedge pressure rises, normally about 6 to 12 mmHg, and it exceeds left ventricular diastolic pressure because the obstruction sits at the valve.

· Anything that shortens diastole makes her worse: exercise, fever, pregnancy, and rapid atrial fibrillation.

· Dyspnea comes from stiff lungs and stimulation of juxtacapillary J receptors, and lying flat shifts blood into the thorax, giving orthopnea.

· The chest film shows Kerley B lines and upper lobe vessel prominence, and chronic congestion leaves hemosiderin-laden macrophages; flooded alveoli give basal crackles and act as shunt.

· In massive pulmonary embolism, clot obstructs the bed and platelet thromboxane A2 and serotonin constrict the rest.

· A thin unhypertrophied right ventricle cannot generate a mean pressure much above about 40 mmHg, and obstruction of about 60% or more of the circulation typically causes acute failure or sudden death.

Distinctions that matter

· Cor pulmonale is defined by where the pulmonary hypertension came from, not by the appearance of the failing right ventricle.

· Wedge pressure separates group 2 from the rest: it is raised when the left heart is the source and normal when the lung is.

Common misconceptions

· Calling any right ventricular failure with raised venous pressure cor pulmonale. Left heart disease is excluded by definition.

· Expecting a previously normal right ventricle to generate high pressures acutely. Hypertrophy takes time, which is why sudden obstruction is so poorly tolerated.

Where it is taught: Section C04, cor pulmonale paragraphs, the pulmonary hypertension group table, the left heart failure paragraph, and the stop and predict box of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C04-PR01

Prediction exercise: Predict the course of a massive pulmonary embolus

A competent response contains

· Right ventricular afterload rises, from mechanical obstruction by clot plus constriction of the remaining bed by platelet thromboxane A2 and serotonin.

· The right ventricle dilates and its output falls, because a thin unhypertrophied wall cannot generate a mean pressure much above about 40 mmHg.

· The septum shifts into the left ventricle, since the pericardium limits the space the two ventricles share.

· Left ventricular filling falls, so left ventricular output and systemic arterial pressure fall.

· Right coronary perfusion falls with arterial pressure while right ventricular wall stress stays high, so supply drops as demand rises.

· Right ventricular ischemia weakens contraction further, closing the spiral, which is why hypotension in pulmonary embolism signals high mortality.

· Arterial PO_2 falls while arterial PCO_2 is normal or low: the blocked region becomes dead space, diverted blood overperfuses the rest and creates low ventilation-perfusion regions, atelectasis and sometimes a right-to-left shunt through a patent foramen ovale add to the hypoxemia, and hyperventilation clears the extra carbon dioxide.

Distinctions that matter

· Obstruction and vasoconstriction both raise afterload, and the constriction is why the hemodynamic effect can exceed what the clot volume alone suggests.

· The hypoxemia here is mainly low ventilation-perfusion rather than true shunt, which is why it usually responds to oxygen.

Common misconceptions

· Expecting a raised arterial PCO2 because a large part of the lung has become dead space. Chemoreceptor-driven hyperventilation keeps it normal or low.

· Treating the fall in left ventricular output as left ventricular failure. The left ventricle is underfilled, not weak.

Where it is taught: Section C04, massive embolism paragraph and the right ventricular failure spiral, with the hypoxemia of embolism in section C05 of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C05-BD1

Brain dump 1 · foundational retrieval: Draw the upright lung and its three zones

A competent response contains

· Pressure rises about 1 cmH2O per centimetre of depth, and 1 mmHg is about 1.36 cmH2O.

· The apex sits about 20 cm above the pulmonic valve and the base about 10 cm below it.

· With a pulmonary artery pressure of 25/8 mmHg, apical pressure is about 10 mmHg in systole and below zero in diastole, while basal pressure is about 33/16 mmHg.

· Blood flow per unit volume therefore rises steeply from apex to base.

· Capillaries in the alveolar walls are also exposed to alveolar pressure, which stays near atmospheric in quiet breathing, and comparing arterial, alveolar, and venous pressure defines the three zones.

· Zone 1: alveolar greater than arterial greater than venous, no flow, no driving pressure, alveolar dead space, normally absent and seen at the apex only with low arterial or high alveolar pressure.

· Zone 2: arterial greater than alveolar greater than venous, intermittent flow mainly in systole, driving pressure is arterial minus alveolar, in the upper and middle lung.

· Zone 3: arterial greater than venous greater than alveolar, continuous flow, driving pressure is arterial minus venous, in the lower lung.

· In zone 2 the outflow end of the capillary collapses to alveolar pressure, so venous pressure does not affect flow, like a waterfall whose flow does not depend on the drop below it.

· Zone 1 is normally absent because apical systolic pressure, about 10 mmHg, still exceeds alveolar pressure.

· Moving down the lung, arterial and venous pressure rise with depth while alveolar pressure stays constant.

Distinctions that matter

· The zones are defined by the order of three pressures, not by fixed anatomical levels, so they move when any of the three changes.

· Zone 1 wastes ventilation while a shunt wastes perfusion; they are opposite failures of matching.

Common misconceptions

· Treating the zones as fixed thirds of the lung. The boundaries shift with posture, arterial pressure, and alveolar pressure.

· Using arterial minus venous pressure as the driving pressure in zone 2. The relevant downstream pressure there is alveolar.

Where it is taught: Section C05, first paragraphs and the table of zones of the upright lung of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C05-BD2

Brain dump 2 · integration and application: Make zone 1 appear, then work the ventilation and perfusion ratios

A competent response contains

· Zone 1 appears when pulmonary artery pressure falls, as in hemorrhage or shock, or when alveolar pressure rises, as in positive pressure ventilation with positive end-expiratory pressure.

· A positive end-expiratory pressure of 15 cmH2O is about 11 mmHg, which exceeds an apical systolic pressure of about 10 mmHg, so apical capillaries carry no flow.

· Zone 1 alveoli are dead space, so arterial PCO2 rises unless ventilation increases.

· Exercise does the reverse, recruiting apical capillaries and converting zone 2 to zone 3.

· Pleural pressure is about -10 cmH2O at the apex and about -2.5 cmH2O at the base at functional residual capacity, because of the weight of the lung.

· Apical alveoli are therefore more expanded and sit on the flat, less compliant part of the pressure-volume curve, so they take in less of each breath, while basal alveoli expand more.

· Ventilation rises toward the base, but less steeply than perfusion does.

· Average ventilation-perfusion ratio is about 0.8, from alveolar ventilation of about 4 L/min over a cardiac output of about 5 L/min.

· At the apex the ratio is about 3, with alveolar PO2 about 130 mmHg and PCO2 about 28 mmHg; at the base it is about 0.6, with PO2 about 90 mmHg and PCO2 about 42 mmHg.

· The high apical PO2 is the usual explanation for the upper lobe location of reactivation tuberculosis, since Mycobacterium tuberculosis is an obligate aerobe.

· High ratio units cannot make up for low ratio units in oxygen because their blood is already nearly saturated on the flat top of the dissociation curve, and the apex also receives little flow.

· They can unload extra carbon dioxide, because the carbon dioxide curve is nearly linear, so mismatch gives hypoxemia with a normal or low arterial PCO2.

· Normal regional mismatch plus the small anatomical shunt produce the normal alveolar to arterial PO2 difference of about 5 to 15 mmHg in young adults.

· Supine, the apex to base differences largely disappear and dependent posterior regions receive more flow.

Distinctions that matter

· Oxygen and carbon dioxide behave differently in mismatch because of the shape of their dissociation curves, which is why hypoxemia appears while PCO2 stays normal or low.

· Apical alveoli are the most expanded but the least well ventilated per breath, since compliance and not resting volume sets the share of each breath they take.

Common misconceptions

· Expecting hyperventilation of healthy lung to correct the hypoxemia of mismatch. Blood leaving those units is already nearly fully saturated.

· Reading a high apical ventilation-perfusion ratio as good gas exchange. The apex contributes little to arterial blood because it receives little flow.

Where it is taught: Section C05, paragraphs on zone 1, on ventilation and ratios from apex to base, and on the supine lung of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C06-BD1

Brain dump 1 · foundational retrieval: Shunt, and the work the pulmonary endothelium does

A competent response contains

· A shunt is blood that reaches the systemic arteries without passing ventilated alveoli, a ventilation-perfusion ratio of zero.

· Anatomical shunts: right-to-left cardiac connections, pulmonary arteriovenous malformations, and the normal 2% from bronchial and Thebesian veins.

· Intrapulmonary shunts are perfused alveoli that are collapsed or filled, as in atelectasis, lobar pneumonia, pulmonary edema, and ARDS.

· Angiotensin-converting enzyme, a zinc peptidase abundant in the pulmonary capillaries and also called kininase II, removes a terminal dipeptide from angiotensin I to form angiotensin II.

· The same enzyme inactivates bradykinin, so blocking it lets bradykinin accumulate, which explains the dry cough and the angioedema of ACE inhibition.

· In one passage the lung removes much of the serotonin, prostaglandins E2 and F2 alpha, and leukotrienes, and about 30% of norepinephrine.

· Epinephrine, angiotensin II, and vasopressin pass through largely unchanged.

· The capillaries filter small venous clots, fat, and air.

· Small emboli rarely infarct because bronchial arteries keep the tissue alive and endothelial fibrinolysis dissolves clots; infarction is more likely when bronchial flow is also poor, as in left heart failure.

· A clot crossing a patent foramen ovale, present in about a quarter of adults, is a paradoxical embolism.

Distinctions that matter

· Anatomical shunt bypasses the alveoli entirely; intrapulmonary shunt perfuses alveoli that are present but not ventilated.

· The lung clears some mediators in a single pass and leaves others untouched, so the pulmonary endothelium is selective rather than a general filter.

Common misconceptions

· Treating all shunt as congenital or cardiac. Lobar pneumonia and alveolar edema are common causes.

· Attributing the cough of ACE inhibition to a lack of angiotensin II. It follows accumulation of bradykinin, which the same enzyme normally destroys.

Where it is taught: Section C06, first paragraph, the nonrespiratory functions paragraphs, and the final paragraph on filtration of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C06-BD2

Brain dump 2 · integration and application: Separate the causes of hypoxaemia with oxygen and arithmetic

A competent response contains

· High altitude: low inspired PO2, normal alveolar to arterial difference, PCO2 low from hyperventilation, corrected by 100% oxygen.

· Opioid overdose: hypoventilation, normal alveolar to arterial difference with a high PCO2, corrected by 100% oxygen.

· Interstitial fibrosis: diffusion limitation, raised difference that worsens with exercise, PCO2 normal or low, corrected by 100% oxygen.

· COPD exacerbation: low ventilation-perfusion regions, raised difference, PCO2 variable and often raised in severe disease, corrected by 100% oxygen.

· Dense lobar pneumonia: shunt with a ratio of zero, raised difference, PCO2 normal or low, little improvement on 100% oxygen.

· Shunted blood never meets alveolar gas, and blood from ventilated alveoli is already about 97 to 98% saturated, so raising its PO2 adds only dissolved oxygen at 0.003 mL/dL per mmHg.

· Mixing combines oxygen content rather than partial pressure, and the steep part of the dissociation curve turns the diluted content into a low PO2.

· On room air, ventilated blood carries about 19.9 mL/dL, so arterial content is 0.7 times 19.9 plus 0.3 times (content minus 5), which gives about 17.8 mL/dL, a saturation near 88% and a PaO2 near 55 mmHg.

· On 100% oxygen with an alveolar PO2 of about 660 mmHg, ventilated blood carries 20.1 bound plus about 2.0 dissolved, or 22.1 mL/dL, so arterial content rises only to about 20.0 mL/dL and PaO2 to about 100 mmHg, against more than 500 mmHg in a normal lung.

· Arterial PCO2 stays normal or low because chemoreceptor-driven hyperventilation unloads the extra carbon dioxide.

· A left-to-right shunt causes no cyanosis at first because the flow direction keeps oxygenated blood in the pulmonary circuit.

· The extra pulmonary flow, and in a ventricular defect or a patent ductus the systemic-level pressure, drives endothelial injury, medial hypertrophy, and intimal fibrosis.

· The six steps: the shunt raises pulmonary flow and often pressure; endothelial injury, medial hypertrophy, and intimal fibrosis follow; pulmonary vascular resistance and right-sided pressures rise progressively; right-sided pressures exceed left-sided pressures; the shunt reverses to right-to-left; late cyanosis, clubbing, and polycythemia appear.

· That end state is Eisenmenger syndrome, and by then the remodeled vessels no longer relax.

Distinctions that matter

· Only shunt resists 100% oxygen; low ventilation-perfusion regions respond, which is why oxygen helps in COPD and embolism but not in dense consolidation.

· Alveolar dead space wastes ventilation and raises PCO2, while shunt wastes perfusion and lowers PO2.

Common misconceptions

· Averaging partial pressures when two streams of blood mix. Content mixes, and the dissociation curve then sets the resulting partial pressure.

· Expecting a shunt to raise PCO2 because blood bypasses the alveoli. Hyperventilation clears carbon dioxide easily because its curve is nearly linear.

Where it is taught: Section C06, paragraph on 100% oxygen, the worked shunt example, the hypoxemia table, and the Eisenmenger flow diagram of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C06-CP02

Comparison table: Mechanisms of arterial hypoxaemia

A competent response contains

· Low inspired PO2: difference normal, PCO2 low from hyperventilation, corrected by oxygen, example high altitude.

· Hypoventilation: difference normal, PCO2 high, corrected by oxygen, example opioid overdose.

· Diffusion limitation: difference increased and worse with exercise, PCO2 normal or low, corrected by oxygen, example interstitial fibrosis.

· Low ventilation-perfusion regions: difference increased, PCO2 normal or low and sometimes raised in severe airflow obstruction, corrected by oxygen, examples COPD, asthma, pulmonary embolism.

· Shunt with a ratio of zero: difference increased, PCO2 normal or low, little improvement with oxygen, examples right-to-left cardiac shunt, lobar pneumonia, alveolar edema.

· Alveolar dead space with a very high ratio: ventilation is wasted rather than perfusion, so PCO2 rises unless ventilation increases, example zone 1 lung created by positive end-expiratory pressure.

Distinctions that matter

· The two mechanisms with a normal alveolar to arterial difference are low inspired oxygen and hypoventilation, and the PCO2 separates those two.

· The response to 100% oxygen separates shunt from everything else.

Common misconceptions

· Expecting the alveolar to arterial difference to be raised in hypoventilation. The alveolar PO2 is low, and the blood follows it.

· Grouping dead space with shunt. Dead space wastes ventilation and raises PCO2, the mirror image of shunt.

Where it is taught: Section C06, table on mechanisms of arterial hypoxemia, with dead space in section C05 of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C06-CA01

Clinical application: Hypoxaemia that does not lift on oxygen

A competent response contains

· Recognize: shunt hypoxemia from dense right lower lobe consolidation, with a ventilation-perfusion ratio of zero in the affected units; the two pointers are the PaO2 rising only from 54 to 71 mmHg on close to 100% oxygen and the raised alveolar to arterial difference alongside a low PaCO2.

· Explain: shunted blood never meets alveolar gas, and blood from the ventilated left lung is already about 97 to 98% saturated, so extra alveolar oxygen adds only dissolved oxygen at 0.003 mL/dL per mmHg; the two streams mix by content, and because the mixed content sits on the steep part of the dissociation curve the resulting PO2 stays low.

· Differentiate: in opioid overdose the alveolar to arterial difference is normal and the PaCO2 is high, whereas his difference is raised and his PaCO2 is 31 mmHg; in a COPD exacerbation the hypoxemia comes from low ventilation-perfusion regions rather than a ratio of zero, so oxygen corrects it, while his PaO2 barely moved.

· Investigate: a wedge pressure of 9 mmHg in the normal range of about 6 to 12 mmHg and a mean pulmonary artery pressure of 18 mmHg exclude a raised left atrial pressure; a normal heart size with unilateral lobar consolidation and a clear left lung fits a lung source; the failure of PaO2 to rise on 100% oxygen measures the shunt itself.

· Decide: hypoxic pulmonary vasoconstriction is constricting the arterioles of the consolidated lobe and diverting blood to the ventilated left lung, and without that response more blood would pass through unventilated alveoli and his PaO2 would be lower still.

· Justify: the stimulus is the low alveolar PO2 of the consolidated units, which becomes strong below about 70 mmHg and acts locally by diffusing to the small arteries; hypoxia inhibits Kv channels, the membrane depolarizes, L-type Ca2+ channels open, and calcium with Rho-kinase drives contraction; mean pulmonary artery pressure stays near normal because the hypoxia is regional and the rest of the bed recruits closed capillaries and distends open ones to absorb the diverted flow.

Distinctions that matter

· A normal wedge pressure with severe hypoxemia places the problem in the lung rather than the left heart.

· Regional hypoxia protects arterial oxygen with little pressure cost, while global hypoxia raises pulmonary artery pressure and gains nothing.

Common misconceptions

· Reading the small rise in PaO2 as inadequate oxygen delivery. It is the expected behaviour of a shunt and it is how a shunt is identified.

· Expecting a whole constricted lobe to produce pulmonary hypertension. Recruitment and distension in the remaining bed keep pressure near normal.

Where it is taught: Section C06, shunt paragraphs, worked example, and hypoxemia table, with regional hypoxia in C02 and recruitment in C01 of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C01-Q01

Board-style question: Why pulmonary pressure barely rises in exercise

Correct answer: C. Opening of previously unperfused capillaries together with widening of those already open

Why it is right: A small rise in pulmonary artery pressure opens capillaries that were closed or unperfused at rest, adding parallel channels, and widens those already open, and since resistance varies inversely with the fourth power of the radius this lowers resistance enough to let flow triple with a rise in mean pressure of only about 7 mmHg.

A is wrong: Pulmonary arterioles have little smooth muscle and little resting sympathetic tone to withdraw, and exercise raises rather than lowers sympathetic activity.

B is wrong: The bronchial circulation carries only about 1 to 2% of cardiac output at systemic pressure and supplies the airways, so it cannot absorb a 10 L/min rise in output.

D is wrong: A rise in alveolar pressure compresses alveolar wall capillaries and would raise pulmonary vascular resistance rather than lower it.

A competent response contains

· Pulmonary vascular resistance fell from 2 Wood units at rest to 0.8 Wood units in exercise, because flow tripled while the pressure drop rose only from 10 to 12 mmHg.

· The fall is passive: recruitment of closed or unperfused capillaries, mostly apical, and distension of those already open.

Distinctions that matter

· Recruitment dominates at low pressure and distension at higher pressure, and both are responses to pressure rather than active dilation.

Common misconceptions

· Looking for a vasodilator mechanism in a bed whose arterioles have very little smooth muscle.

Where it is taught: Section C01 of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C04-Q02

Board-style question: Alveolar flooding at a normal wedge pressure

Correct answer: A. Injury to the alveolar capillary barrier that allows protein to cross at a normal capillary pressure

Why it is right: Sepsis injures the alveolar capillary barrier so protein crosses into the interstitium and alveoli, which removes the inward oncotic force that normally holds fluid in, and edema forms even though the capillary hydrostatic pressure and the wedge pressure are normal.

B is wrong: Backward transmission of a raised left atrial pressure would raise the wedge pressure above 12 mmHg and usually enlarge the heart, and the fluid would be protein poor.

C is wrong: A low plasma oncotic pressure narrows the safety margin but still needs a rise in capillary pressure to flood the alveoli, and it does not let plasma protein cross into the airway.

D is wrong: Lymphatic drainage is a defence that is overwhelmed in this situation rather than the primary fault, and obstructed lymphatics alone would not produce fluid with the protein content of plasma.

A competent response contains

· A wedge pressure of 8 mmHg is within the normal range of about 6 to 12 mmHg, so left atrial pressure is not raised.

· Edema fluid with a protein concentration close to that of plasma shows the alveolar capillary barrier is leaking protein, which is permeability edema at a normal capillary pressure.

Distinctions that matter

· Hydrostatic edema has a raised wedge pressure and protein-poor fluid; permeability edema has a normal wedge pressure and protein-rich fluid.

Common misconceptions

· Assuming any diffuse alveolar edema means a failing left ventricle. The wedge pressure and the protein content settle it.

Where it is taught: Section C04 of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C05-Q03

Board-style question: Dead space created at the apex

Correct answer: D. Alveolar pressure at the apex now exceeding arterial pressure, so those ventilated alveoli carry no blood flow

Why it is right: Blood loss lowers pulmonary artery pressure and the positive end-expiratory pressure of about 11 mmHg raises alveolar pressure above the apical arterial pressure, creating zone 1 lung, and those ventilated but unperfused alveoli are dead space, so at a fixed minute ventilation the arterial PCO2 climbs.

A is wrong: Hypoxic constriction would redirect blood flow rather than create dead space, and it lowers perfusion to poorly ventilated units, which does not raise PCO2.

B is wrong: Loss of radial traction narrows extra-alveolar vessels at low lung volume, and this patient's lung volume is raised rather than lowered by the applied pressure.

C is wrong: Collapsed alveoli with continued perfusion are shunt, which lowers PO2 and leaves PCO2 normal or low because hyperventilation clears carbon dioxide.

A competent response contains

· A positive end-expiratory pressure of 15 cmH2O is about 11 mmHg, which exceeds the apical systolic pressure produced by a pulmonary artery pressure of 18/6 mmHg over a 20 cm column.

· Alveolar pressure above arterial pressure is zone 1: those alveoli are ventilated but carry no blood flow, so they are dead space and arterial PCO2 rises unless ventilation increases.

Distinctions that matter

· Zone 1 wastes ventilation and raises PCO2, whereas shunt wastes perfusion and lowers PO2.

Common misconceptions

· Looking for a cause of hypoxemia when the measurement that changed is the PCO2. Dead space is a ventilation problem.

Where it is taught: Section C05 of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C03-Q04

Board-style question: A large inspiratory fall in systolic pressure

Correct answer: B. Pericardial fluid fixes the combined ventricular volume, so inspiratory right ventricular filling displaces the septum into the left ventricle

Why it is right: With the pericardium full of fluid the two ventricles share a fixed total volume, so the normal inspiratory rise in right ventricular filling can only be accommodated by displacing the septum into the left ventricle, and left ventricular stroke volume therefore falls far more than the normal few millimetres of mercury.

A is wrong: Large pleural pressure swings explain pulsus paradoxus in severe asthma or COPD, but her lungs are clear and the effusion with equalized diastolic pressures identifies the pericardium as the cause.

C is wrong: Pulmonary arterioles constrict in response to low alveolar oxygen rather than to inspiration, and inspiration expands the pulmonary vessels so they hold more blood rather than constricting.

D is wrong: A rigid pericardium shielding the heart describes constrictive pericarditis, in which pulsus paradoxus is less common and jugular venous pressure rises on inspiration.

A competent response contains

· A fall in systolic pressure of 22 mmHg on inspiration is pulsus paradoxus, an exaggeration of the normal dip of less than 10 mmHg.

· A large circumferential effusion with equal ventricular diastolic pressures is tamponade, in which the pericardial fluid fixes the combined volume of the two ventricles.

Distinctions that matter

· In tamponade the pericardium does the work; in severe airflow obstruction the size of the pleural pressure swing does.

Common misconceptions

· Reading the finding as a reversal of the normal inspiratory change. It is the normal direction, exaggerated.

Where it is taught: Section C03 of the CV-15 Heart and Lung Interactions notes.